Week 2 Tutorial

COMP10001 – Foundations of Computing

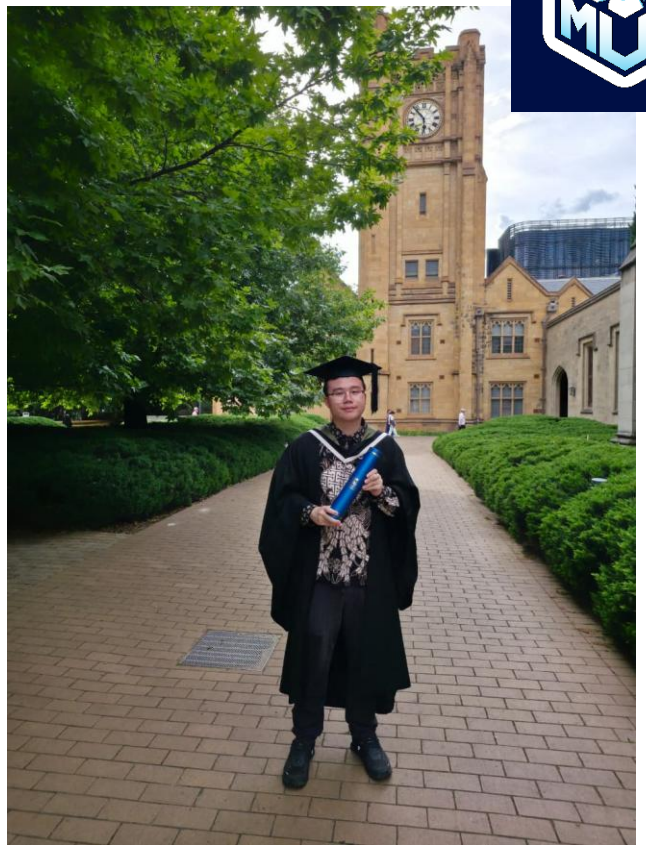
Semester 1, 2026

Clement Chau

- Welcome and subject overview
- Introduction to Computing
- Programming basics (Blockly)

Get to know myself

- Clement Chau (he/him)
- Graduated in December 2025, CSS major
- Went through COMP10001 as a student (2023S1)
- Returned to COMP10001 as a tutor (2025S1, 2025S2)
- Enjoys video games and esports (quite heavily…)
- Was heavily involved in Melbourne University Esports as Head of Technical and Rhythm Games Department Manager (still kinda am!)
- Was once part of the VAL team for the university's esports club.



Get to know yourself

- Name
- Year, Semester, Course
- Choose one (or all) of:
  - Favorite Game
  - Favorite Show/Movie
  - Favorite Place you've visited outside of your city
  - Or any other hobbies that you might want to share to your classmates!

Where can COMP10001 take me?

Software engineer

Building websites, apps

Data scientist/analyst

Amazon recommendations, Google search engine

AI/ML Engineer

Chatbots such as ChatGPT, DeepSeek

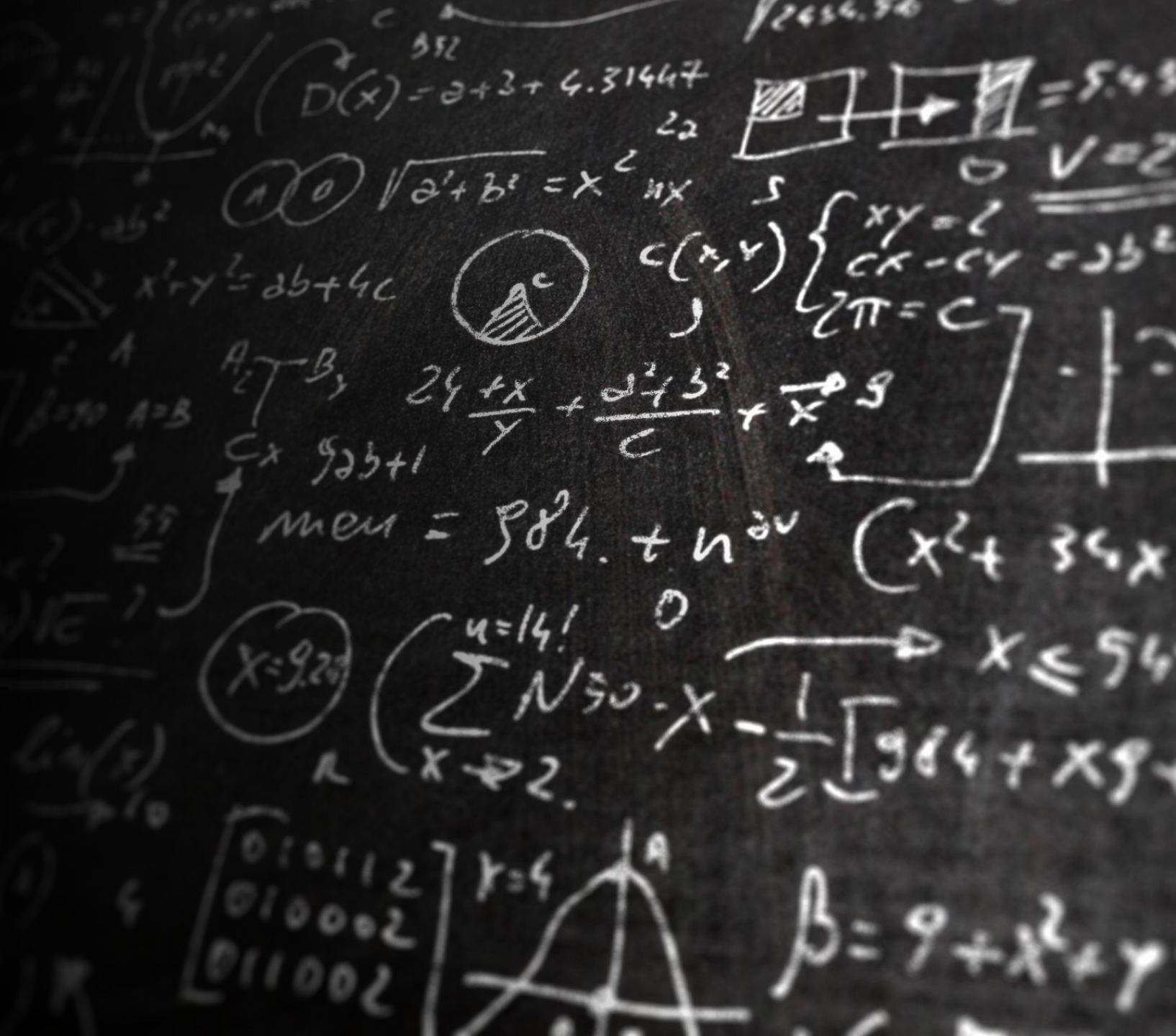
Game Development

Other types of engineering

Electrical Engineering, Robotics

And so much more...

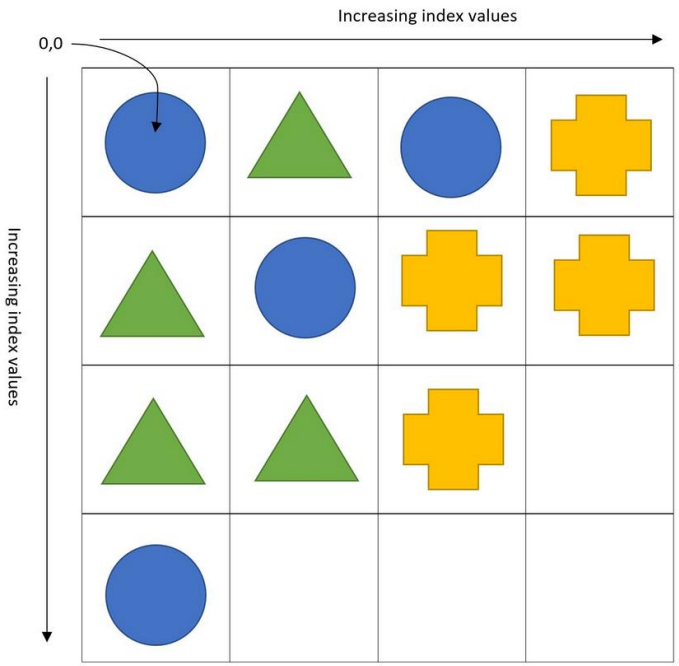
Programming is a very transferrable skill to any field of study and any career



2023S1 projects:

Project 1 – Matching game

```
board = [['B', 'G', 'B', 'Y'],
['G', 'B', 'Y', 'Y'],
['G', 'G', 'Y', 'Z'],
['B', 'Z', 'Z', 'Z']]
```



Our representation

Project 2 – FoCdle

The *difficulty* of a **FoCdle** is measured by its length in total characters.

For example, here is a trace of a person solving a **FoCdle** of difficulty 10. In their first guess they tried the 10-character equation "`13+12-8=17`" and learnt (from the green cells) what the first operator was and where it was located, and got the location of the "=" correct. They also learnt (from the yellow cells) that there were at least one each of the digits 1, 2, 3, and 7 (plus for each of those digits, they learnt one character position in which it did *not* appear); and they learnt (from the grey cells) that there was only a single instance of 1, that the second operator wasn't subtraction, and that there were no 8 digits anywhere.

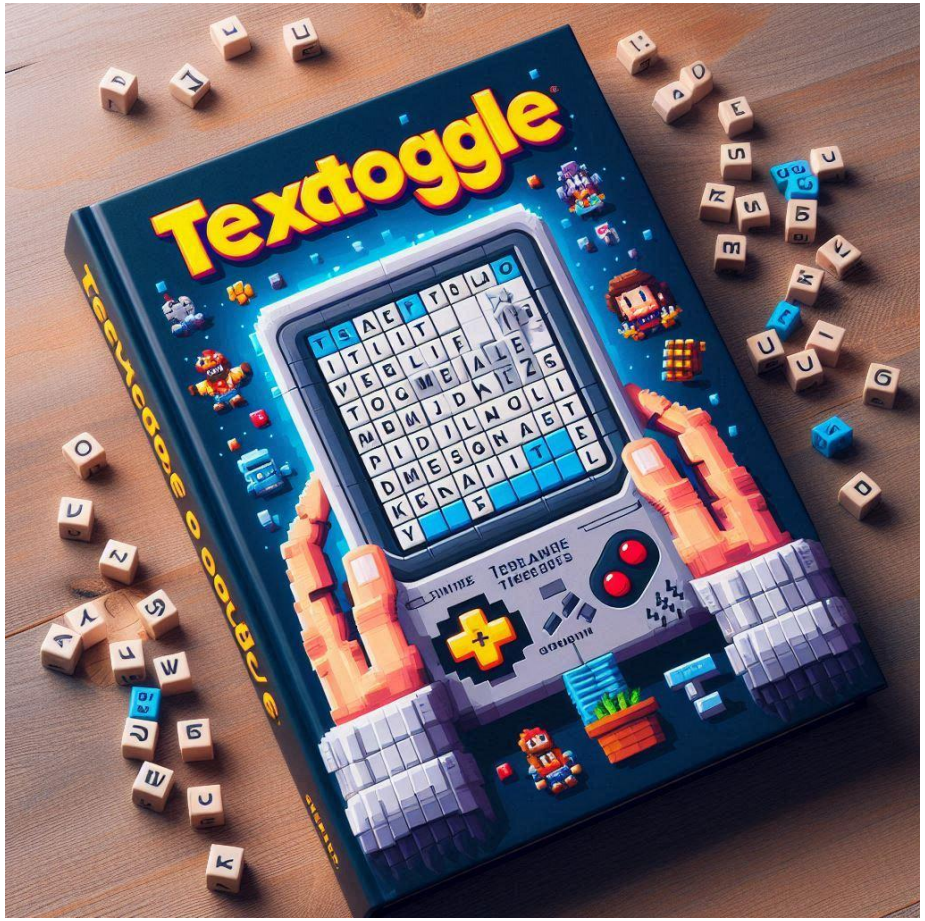
0	1	2	3	4	5	6	7	8	9	
1	3	+	1	2	–	8	=	1	7	guess 1
7	2	+	3	1	%	6	=	7	3	guess 2
2	1	+	2	0	*	4	=	5	5	guess 3
2	5	+	4	*	1	2	=	7	3	guess 4, all green!

Example 1

From that information they formed their second guess "`72+31%6=73`" and submitted it. The response from that told them that the computed value had to be 73; that second operator wasn't "`%`" either; that there were no 6s, only one 7, and only one 3; plus also told them some more positions in which the digits 1 and 2 (which must occur somewhere) could not appear.

2025S1 projects:

Project 1 – Textoggle (Scrabble)



Project 2 – Minescaper (Minesweeper)

Input:

1	💣	1	1	💣
1	1	1	1	1
0	0	0	0	0
0	0	1	1	1
0	0	1	💣	1

Output:

1	💣	1	1	💣
1	1	1	1	1
0	0	0	0	0
0	0	1	1	1
0	0	1	💣	1

visited

current_pos

visited update

Academic Integrity

- Where **NOT** to get help
  - Online
    - Forums, External Tutor
    - You *can* post it on Ed, but make it **PRIVATE!**
  - LLMs (ChatGPT, etc)
  - 'Friends'

In the past we have given penalties such as:

- Zero (0) for the assignment.
- Zero (0) for the subject.
- Termination of enrollment from the University.

IMPORTANT: If in doubt regarding what's allowed and what's not, **ASK THE TEACHING STAFF.**

Getting Help

- Tutorials (2nd hour)
- Student Hub in Level 3 Melbourne Connect
  - Opens 9am to 5pm every day (**bring your student card!!!**)
  - COMP10001 tutors will be there at **11am-1pm on Wednesdays, 3pm-5pm on every other Weekday**
  - https://cis.unimelb.edu.au/study/cis-student-hub
- Ed Discussion forums
- Revisiting lectures and workshop materials in Canvas
  - Consolidation lectures every other Friday

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Bingo time!

Find 3 things that your table has in common

Blocky Turtle!

https://blockly.games/turtle?lang=en&level=10

Can we do this more effectively?

```
set sides to 5
set length to 80
set angle to 180 - 180 ÷ sides
repeat sides times
do move forward by length
   turn right ↻ by angle
```

Can we make it *even* better?

```
to mystery shape with: sides, length
  set angle to (180 - (180 ÷ sides))
  repeat sides times
  do move forward by length
     turn right ↻ by angle
```

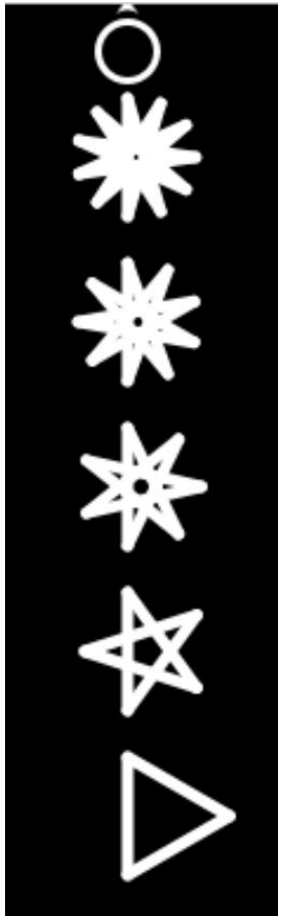
```
mystery shape with:
    sides 5
    length 80
```

Challenge!

```
pen up
move backward by 170
pen down
set sides to 3
repeat 5 times
do  mystery shape with:
        sides sides
        length 50
    pen up
    move forward by 70
    pen down
    change sides by 2
```

What does this draw?

This is a tricky one!



Independent work

- **Do worksheets 1** and **2** Ed **(Due 16 March, 2pm)**
  - Remember that **Ed worksheets contributes to 10% of your total score!**
- **Raise your hand** if you have any questions!